Do Now · retrieval

Lesson 117 · Waterfall and Agile Development

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the difference between linker and loader?

Q2

What is the time complexity of sequential search in the worst case?

Q3

After pushing A then B onto a stack, which is popped first?

Q4

Why must a binary search operate on ordered data?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Waterfall lifecycle — Sequential analysis, design, implementation, testing and maintenance.

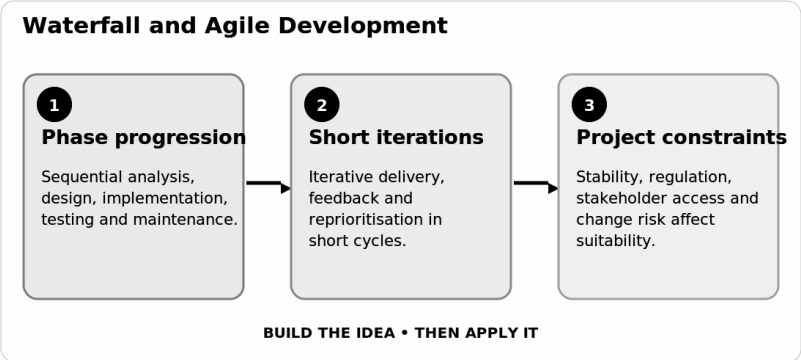
Core idea 2

Agile methods — Iterative delivery, feedback and reprioritisation in short cycles.

Core idea 3

Choose on evidence — Stability, regulation, stakeholder access and change risk affect suitability.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

waterfall — Stage-based plan-driven development lifecycle.

agile — Iterative and adaptive development principles.

iteration — Repeated short cycle yielding feedback/increment.

requirements — Functions/constraints a system must satisfy.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

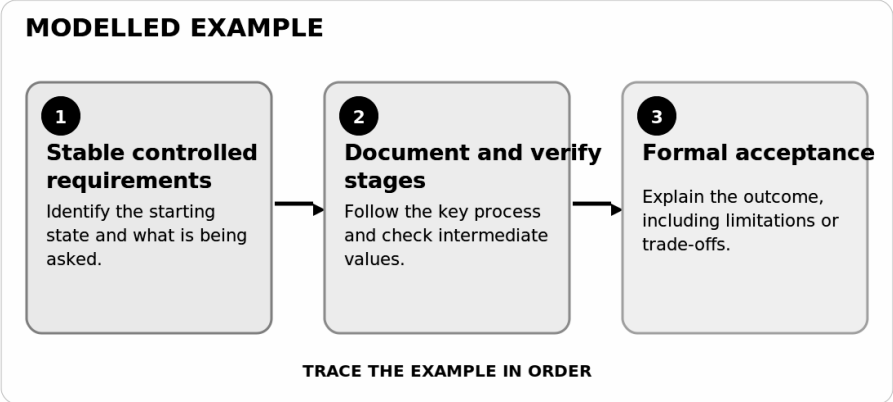
Problem

An air-traffic subsystem has carefully fixed specifications and requires formal approvals. Select a development approach.

Follow the worked method

1. Model answer — A plan-driven/waterfall approach can support documented phase gates and traceability; however, testing and validation must still occur throughout as appropriate. Safety assurance matters more than simply labelling it waterfall. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Describe the waterfall stages and one drawback of late requirement changes. [6]

Question 2

Explain two features of agile development in a public-services app. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate waterfall versus agile for replacing an exam-management system subject to fixed deadlines and changing rules. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Agile means a project needs no testing or documentation.

Use key vocabulary from memory

1. Explain waterfall in your own words.

2. Explain agile in your own words.

3. Explain iteration in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.